

BADMINTON SWING ANALYZER USING MPU6050 WITH ML
MINI-PROJECT REPORT

Submitted by

ASWIN RAAJ R (Reg.no: 202302011)

MANI BALAJI S (Reg.no: 202302056)

THARUN VS (Reg.no: 202302108)

In partial fulfilment for the award of the degree of
BACHELOR OF ENGINEERING IN
ELECTRICAL AND ELECTRONICS ENGINEERING

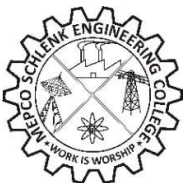
9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



,

MEPCO SCHLENK ENGINEERING COLLEGE,
SIVAKASI

(An Autonomous Institution affiliated to Anna
University Chennai)



MAY 2026

BONAFIDE CERTIFICATE

Certified that this project report titled “**BADMINTON SWING ANALYZER USING MPU6050 WITH ML**” is the bonafide work of **Mr.R.ASWIN RAAJ (REG.No:9517202302011)**, **Mr.S.MANI BALAJI (REG.NO:9517202302056)** and **Mr.V.S.THARUN (REG.NO:9517202302108)** who carried out the mini-project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Dr.G.Pandiaraj, M.E., Ph.D.

Dr. N. Senthil Kumar, M.E., Ph.D.

Supervisor

**Assistant Professor, (Sl.Grade)
Department of Electrical and Electronics
Electronics Engineering,
Mepco Schlenk Engineering College,
Sivakasi.**

Head of the Department

**Senior Professor,
Department of Electrical and
Engineering,
Mepco Schlenk Engineering College,
Sivakasi.**

Submitted for Viva-Voce Examination held at **MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI (AUTONOMOUS)** on

Examiner 1

Examiner 2

ABSTRACT

The rapid advancement of embedded systems and machine learning has enabled the development of intelligent sports analytics solutions. This project presents a Smart Badminton Performance Analytics System designed to monitor, classify, and analyze player performance using real-time motion data. The system utilizes an ESP32 microcontroller integrated with an MPU6050 inertial measurement unit (IMU) to capture acceleration and angular velocity during badminton swings. A threshold-based algorithm is implemented to detect valid swings, and key parameters such as speed, impact, and duration are extracted. These parameters are further processed using feature engineering techniques to derive meaningful metrics such as power, efficiency, and energy. Machine learning models are employed to classify swings into WEAK, MEDIUM, and STRONG categories and to identify stroke types including SMASH, DROP, CLEAR, and DRIVE. A hybrid approach combining machine learning and rule-based logic enhances classification accuracy and system reliability. The processed data is analyzed to generate advanced insights such as player performance level, consistency score, fatigue detection, and comparison with professional benchmarks. The system provides feedback through a web-based dashboard and a Streamlit interface, enabling intuitive visualization of performance metrics. Furthermore, the project outlines a pathway for transitioning from a prototype to a compact wearable product by incorporating ultra-low power microcontrollers such as Seeed Studio XIAO nRF52840 or ESP32-C3, advanced IMU sensors like ICM42688, and lightweight Li-Po battery systems.

ACKNOWLEDGEMENT

An endeavour over a long period can be successful only with the advice and support of many well-wishers. First and foremost, we thank the lord almighty for his abundant grace and blessings for marking this work a great success.

We would like to extend our sincere gratitude to our honourable Principal **Dr.S.ARIVAZHAGAN, M.E., Ph.D., FIETE. MISTE** for providing ample facilities and necessary infrastructure during the course of the project.

We express our regards and thanks to our great support **Dr.N.SENTHIL KUMAR,M.E., Ph.D.,** Senior Professor and Head of Electrical and Electronics Engineering Department for his cordial encouragement throughout the project.

We take great pleasure in expressing our deep sense of gratitude to **Dr.G.PANDIARAJ, M.E., PhD** for his excellent guidance, encouragement and personal interest throughout the course of the project. Without his encouragement and guidance, this project would not have been materialized.

We also thank all the staff members and lab technicians of Electrical and Electronics Engineering Department for their kind co-operation in the completion of the project.

Finally, we express our gratitude to our family and friends for their invaluable assistance throughout this project.

TABLE OF CONTENTS

S.No	Contents	Page.no
1.	Chapter 1: INTRODUCTION	1
2.	Chapter 2: LITERATURE REVIEW	2
3.	Chapter 3: SYSTEM OVERVIEW	5
4.	Chapter 4: HARDWARE COMPONENTS AND SETUP	8
5.	Chapter 5: SOFTWARE ARCHITECTURE	12
6.	Chapter 6: DATA ACQUISITION AND FEATURE ENGINEERING	16
7.	Chapter 7: MACHINE LEARNING AND HYBRID MODEL DESIGN	17
8.	Chapter 8: SYSTEM IMPLEMENTATION AND ALGORITHM DESIGN	18
9.	Chapter 9: RESULTS AND PERFORMANCE ANALYSIS	
10.	Chapter 10: FUTURE AND PRODUCTIZATION	
10.	CONCLUSION	
11.	REFERENCES	

TABLE OF FIGURES

S.No	Figure No.	Figure name	Page No.
1.	4.1	MPU6050	9
2.	4.2	ESP32 Dev Kit V1	9
3.	4.3	Push Button	9
4.	4.4	3.7v Li-Po 1000mAh Battery	9
5.	4.5	TP4056 Charging Module	9
6.	4.6	Badminton Racket	9
7.	9.1	Comprehensive session report	15
8.	9.2		

CHAPTER 1

INTRODUCTION

1.1 Overview

Badminton is a fast-paced sport that requires a combination of speed, precision, timing, and power. Performance analysis in badminton is traditionally done through manual observation by coaches, which is often subjective and lacks quantitative accuracy.

With the advancement of embedded systems, sensors, and machine learning, it is now possible to develop intelligent systems that can capture, analyze, and interpret player performance in real time. This project presents a Smart Badminton Performance Analytics System, which integrates sensor data acquisition, embedded processing, and machine learning techniques to evaluate player swings.

The system uses an inertial measurement unit (IMU) sensor to capture motion data during gameplay. The collected data is processed to extract meaningful features such as swing speed, impact force, and duration. These parameters are further analyzed using machine learning models to classify swing performance and provide insights into player technique.

1.2 Problem Statement

Traditional badminton training methods rely heavily on visual feedback and manual coaching, which lack precision and consistency. Players often do not have access to real-time performance metrics such as swing strength, impact quality, and consistency.

There is a need for a system that can:

- Automatically capture swing data
- Provide real-time performance analysis
- Classify player performance accurately
- Offer data-driven feedback for improvement

1.3 Objectives

The main objectives of this project are:

- To design and develop a sensor-based system for capturing badminton swing data
- To implement real-time swing detection using embedded systems
- To extract meaningful features such as speed, impact, and duration

- To develop machine learning models for swing classification
- To analyze player performance using advanced analytics techniques
- To provide feedback and suggestions for performance improvement

1.4 Scope of the Project

This project focuses on developing a portable and cost-effective system for analyzing badminton player performance. The system is designed to work in real-time and provide insights into swing characteristics.

The scope includes:

- Real-time data acquisition using sensors
- Swing classification into categories (Weak, Medium, Strong)
- Stroke type classification (Smash, Drop, Clear, Drive)
- Performance analysis including consistency, fatigue, and stability
- Visualization of results through a web interface and dashboard

The system can be extended to other sports and enhanced with cloud-based analytics and mobile applications.

1.5 Motivation

The motivation behind this project is to bridge the gap between traditional coaching methods and modern data-driven sports analytics. By leveraging embedded systems and machine learning, the project aims to provide players with objective performance metrics and actionable insights.

This system enables:

- Improved training efficiency
- Better understanding of player strengths and weaknesses
- Data-driven decision making in sports training

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The development of intelligent sports analytics systems has gained significant attention in recent years due to advancements in embedded systems, wearable sensors, and machine learning. In badminton, performance analysis traditionally relies on manual observation, which lacks precision and real-time

feedback. Researchers and developers have explored various approaches to overcome these limitations using technology.

This chapter reviews existing methods and systems related to motion tracking, wearable devices, and machine learning-based sports analytics.

2.2 Wearable Sensor-Based Systems

Wearable devices equipped with inertial sensors such as accelerometers and gyroscopes are widely used for motion tracking in sports. These systems are capable of capturing real-time movement data and are commonly used in fitness tracking and activity recognition.

In badminton and similar sports, inertial measurement units (IMUs) are used to capture swing motion. These sensors provide raw data such as acceleration and angular velocity, which can be processed to extract meaningful performance metrics.

Limitations:

- Limited analysis capabilities in basic systems. Lack of advanced feature extraction. Minimal or no real-time feedback

2.3 Vision-Based Motion Analysis Systems

Camera-based systems use computer vision techniques to analyze player movements. These systems track body posture, racket movement, and shuttle trajectory using image processing algorithms.

Drawbacks:

- High computational requirements. Dependency on lighting and environment. Not portable

2.4 Machine Learning in Sports Analytics

Machine learning has been widely applied in sports analytics to classify player actions and predict performance. Algorithms such as decision trees, support vector machines, and neural networks are used to analyze sensor data.

In motion-based sports:

- Classification models identify movement patterns. Feature engineering improves prediction accuracy. Data-driven insights help in performance optimization. Focus only on classification. Do not integrate real-time embedded systems. Lack comprehensive analytics (fatigue, consistency, etc.)

2.5 Existing Badminton Analysis Systems

Some existing badminton analysis systems include. Smart rackets with built-in sensors. Mobile applications for swing tracking. Basic motion analysis tools

These systems provide limited features such as, Swing speed measurement. Basic shot detection

Limitations are High cost. Limited analytics capabilities. No personalized coaching insights. Lack of integrated AI-based feedback

2.6 Identified Research Gap

From the study of existing systems, the following gaps are identified:

- Lack of integrated systems combining embedded hardware and machine learning. Limited real-time analytics capabilities. Absence of advanced performance metrics such as:
 - Consistency
 - Stability
 - Fatigue detection
- No comprehensive player performance evaluation
- Lack of intelligent feedback or coaching suggestions

2.7 Proposed System Contribution

To address these limitations, the proposed system introduces:

- A sensor-based embedded system for real-time data acquisition
- A machine learning-based classification system for swing and stroke analysis
- A hybrid approach combining rule-based and ML models
- Advanced analytics including:
 - Player vs professional comparison
 - Consistency and stability analysis
 - Fatigue detection
- A user-friendly interface for real-time monitoring and data visualization

This approach provides a cost-effective, portable, and intelligent solution for badminton performance analysis.

CHAPTER 3

SYSTEM OVERVIEW

3.1 Introduction

The Smart Badminton Performance Analytics System is designed as an integrated platform that combines embedded hardware, data processing, machine learning, and user interface modules to analyze player performance in real time. The system captures motion data from the player's racket, processes it to extract meaningful features, and applies intelligent algorithms to evaluate performance.

This chapter presents the overall architecture and working pipeline of the proposed system.

3.2 Overall System Architecture

The system consists of multiple interconnected layers that work together to perform data acquisition, processing, analysis, and visualization.

Main Components:

- Sensor Layer (MPU6050)
- Embedded Processing Layer (ESP32)
- Data Processing & Feature Engineering Layer
- Machine Learning Layer
- Analytics & Decision Layer
- User Interface Layer

System Flow:

Motion → Sensor Data → ESP32 Processing → Feature Extraction → Data Storage (CSV) → ML Models → Analytics Engine → User Interface

3.3 Working Pipeline

The complete working of the system is divided into the following stages:

3.3.1 Data Acquisition

- Motion data is captured using the MPU6050 sensor
- Accelerometer and gyroscope provide raw values:
 - Ax, Ay, Az, Gx, Gy, Gz
- ESP32 reads sensor data continuously using I2C communication

3.3.2 Signal Processing

- Raw data is calibrated to remove sensor offsets
- Gravity component is filtered using a smoothing algorithm
- Total acceleration and angular velocity are calculated

These processed values form the basis for further analysis.

3.3.3 Swing Detection

The system identifies a valid swing using threshold-based logic:

- Start Condition: Acceleration exceeds a predefined threshold
- Tracking Phase: Peak values of speed and impact are recorded
- End Condition: Acceleration drops below threshold and minimum duration is satisfied

For each detected swing, the following parameters are extracted:

- Peak speed
- Maximum impact
- Swing duration

3.3.4 Data Logging

- Each swing is stored in CSV format
- Data includes:
 - Sensor readings
 - Computed features
 - Timestamp
- Data can be downloaded via the web interface for further analysis

3.3.5 Feature Engineering

The collected data is processed to generate meaningful features:

- $\text{Power} = \text{Speed} \times \text{Impact}$
- $\text{Efficiency} = \text{Impact} / \text{Duration}$

These features improve the performance of machine learning models and enable deeper analysis.

3.3.6 Machine Learning Processing

The system uses two machine learning models:

1. Swing Classification Model
 - Classifies swings into:
 - Weak, Medium, Strong
2. Stroke Classification Model
 - Identifies stroke type:
 - Smash, Drop, Clear, Drive

Additionally, a rule-based threshold system is used alongside ML predictions to improve reliability.

3.3.7 Analytics and Decision Making

An advanced analytics engine processes the classified data to generate insights:

- Player vs professional comparison
- Gap analysis
- Consistency and stability evaluation
- Fatigue detection
- Player type classification

Based on these analyses, the system generates performance feedback and improvement suggestions.

3.3.8 User Interface

The system provides multiple interfaces:

- ESP32 Web Dashboard
 - Displays real-time speed, impact, and swing count
- Terminal Dashboard
 - Shows detailed analytics results
- Streamlit Application
 - Provides interactive visualization and insights

3.4 System Modes of Operation

The system operates in different modes:

- Calibration Mode
 - Used to collect initial data and set thresholds
- Game Mode
 - Used for real-time performance tracking
- Idle Mode
 - System remains inactive

3.5 Advantages of the Proposed System

- Integration of embedded systems and machine learning, combining hardware sensing with intelligent analysis
- Hybrid approach (machine learning + rule-based logic) for improved accuracy and reliability
- Advanced analytics beyond basic classification, including consistency, stability, and fatigue analysis
- Custom calibration mechanism for personalized threshold setting based on player data
- Coaching and feedback system for performance improvement

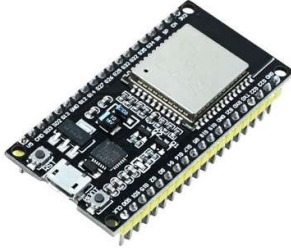
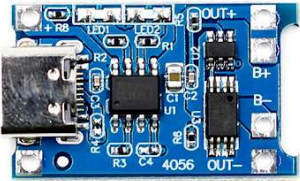
CHAPTER 4

HARDWARE COMPONENTS AND SETUP

4.1 Introduction

The hardware layer forms the foundation of the proposed system. It is responsible for capturing real-time motion data and enabling interaction with the user. The system primarily consists of a microcontroller, motion sensor, input controls, and output indicators.

This chapter describes the components used, their working principles, and the overall hardware setup.

Components	Components
 Fig 4.1 MPU6050	 Fig 4.2 ESP32 Dev Kit V1
 Fig 4.3 Push Button	 Fig 4.4. Li-po 3.7v 1000mah battery
 Fig 4.5 TP4056 Charging Module	 Fig 4.6 Badminton Racket.

4.2 ESP32 Microcontroller

The ESP32 is the main processing unit of the system. It is a powerful, low-cost microcontroller with built-in Wi-Fi capabilities, making it suitable for IoT-based applications.

Key Features:

- Dual-core processor
- Built-in Wi-Fi and Bluetooth
- High processing speed
- Multiple GPIO pins
- Supports I2C communication

Role in the System:

- Reads sensor data from MPU6050
- Performs real-time signal processing
- Detects swings using threshold logic
- Hosts a web server for user interface
- Handles data storage and communication

4.3 MPU6050 Sensor (IMU)

The MPU6050 is a 6-axis Inertial Measurement Unit (IMU) that combines an accelerometer and a gyroscope.

Working Principle:

- Accelerometer: Measures linear acceleration along X, Y, and Z axes
- Gyroscope: Measures angular velocity along X, Y, and Z axes

Specifications:

- 3-axis accelerometer + 3-axis gyroscope
- Digital output via I2C protocol
- High sensitivity and accuracy

Role in the System:

- Captures motion data during racket swings
- Provides raw data for feature extraction

- Enables calculation of speed, impact, and motion patterns

4.4 Input Components

Push Buttons

Two push buttons are used to control system modes:

- Calibration Button
 - Used to start/stop calibration mode
 - Helps collect initial dataset
- Game Button
 - Used to start/stop game mode
 - Enables real-time performance tracking

4.5 Output Components

LED Indicator

- Turns ON when a swing is detected
- Turns OFF when the swing ends

This helps in debugging and real-time system verification.

4.6 Power Supply

The system is powered using a stable DC power source such as:

- USB power supply
- Battery pack

The ESP32 regulates voltage internally to support connected components.

4.7 Hardware Interfacing

MPU6050 to ESP32 Connection (I2C):

MPU6050 Pin	ESP32 Pin
VCC	3.3V
GND	GND

MPU6050 Pin	ESP32 Pin
SDA,SCL	GPIO 26 GPIP 25

Other Connections:

Component	ESP32 Pin
LED	GPIO 18,19
Calibration Button	GPIO 22
Game Button	GPIO 23

4.8 Circuit Description

- The MPU6050 communicates with ESP32 via I2C protocol
- Buttons are connected using pull-up configuration
- LED is connected as a digital output indicator
- ESP32 processes all inputs and controls outputs

4.9 Working of Hardware Setup

1. The MPU6050 continuously senses motion
2. ESP32 reads sensor data in real time
3. Buttons allow switching between modes
4. LED provides visual indication of swing detection
5. ESP32 processes and sends data to the web interface

4.10 Advantages of Hardware Design

- Compact and portable setup
- Low power consumption
- Cost-effective implementation
- Easy integration with software modules
- Reliable real-time data acquisition

CHAPTER 5

SOFTWARE ARCHITECTURE

5.1 Introduction

The software architecture of the Smart Badminton Performance Analytics System integrates embedded firmware, data processing, machine learning models, and user interface modules. The system is designed as a multi-layer architecture where each layer performs a specific function and communicates with other layers to achieve real-time performance analysis.

5.2 Overall Software Architecture

The system is divided into the following layers:

- Embedded Firmware Layer (ESP32)
- Data Processing Layer (Python)
- Machine Learning Layer
- Analytics & Decision Layer
- User Interface Layer (Web + Streamlit)

Software Flow:

ESP32 → CSV Data → Python Processing → ML Models → Analytics → Dashboard

5.3 Embedded Firmware (ESP32)

The ESP32 firmware is responsible for real-time data acquisition, preprocessing, swing detection, and communication.

Core Functionalities:

- Sensor initialization (MPU6050)
- Continuous data acquisition
- Signal filtering and processing
- Swing detection using thresholds
- Data logging (CSV format)
- Hosting web server for monitoring

5.3 Data Processing & Feature Engineering

After data collection, Python is used to process and enhance the dataset.

Key Features Created:

- $\text{Power} = \text{Speed} \times \text{Impact}$
- $\text{Efficiency} = \text{Impact} / \text{Duration}$
- $\text{Energy} = \text{Power} \times \text{Duration}$

Pseudo Code: Feature Engineering

FOR each swing in dataset:

```
power = speed * impact  
efficiency = impact / duration  
energy = power * duration
```

Code Concept Snippet

```
df["power"] = df["speed"] * df["impact"]  
df["efficiency"] = df["impact"] / (df["duration"] + 1e-6)
```

5.4 Machine Learning Layer

The system uses two trained machine learning models:

1. Swing Classification Model

- Input: engineered features
- Output: WEAK / MEDIUM / STRONG

2. Stroke Classification Model

- Input: advanced motion features
- Output: SMASH / DROP / CLEAR / DRIVE

Pseudo Code: ML Prediction

```
LOAD swing_model  
LOAD stroke_model
```

FOR each swing:

```
swing_class = swing_model.predict(features)  
stroke_type = stroke_model.predict(features)
```

5.5 Analytics & Decision Engine

This module performs advanced analysis on classified data.

Functions Implemented:

- Player vs Pro comparison, Gap analysis, Consistency calculation, Stability scoring, Fatigue detection, Player type classification, AI suggestions

Pseudo Code: Analytics

COMPUTE average speed, impact, power

COMPARE with professional dataset

CALCULATE:

gap values

consistency (std deviation)

fatigue (early vs late swings)

GENERATE suggestions based on gaps

1. results
2. UI displays final insights

CHAPTER 6

DATA ACQUISITION AND FEATURE ENGINEERING

6.1 Introduction

Data acquisition and feature engineering play a critical role in the performance of the Smart Badminton Analytics System. The quality of data and the relevance of extracted features directly influence the accuracy of classification and analytics. This chapter describes the process of collecting motion data, structuring datasets, and transforming raw sensor readings into meaningful features for machine learning and analysis.

6.2 Data Acquisition

The system collects real-time motion data using the MPU6050 sensor integrated with the ESP32 microcontroller.

Captured Parameters:

- Accelerometer data: Ax, Ay, Az
- Gyroscope data: Gx, Gy, Gz

- Timestamp

These values are continuously read and processed to detect valid badminton swings.

6.3 Swing-Based Data Collection

Instead of storing continuous raw data, the system captures event-based swing data, which improves efficiency and relevance.

Each swing is detected using threshold-based logic and the following parameters are recorded:

- Peak speed
- Maximum impact
- Swing duration
- Processed sensor values

6.4 Dataset Structure

The collected data is stored in CSV format for further processing.

Sample Dataset Format:

timestamp, ax, ay, az, gx, gy, gz, speed, impact, duration

Each row represents a single detected swing.

6.5 Calibration Dataset and Thresholding

A calibration phase is used to establish baseline performance metrics.

Process:

- First set of swings (~30 samples) is collected
- Features are computed for calibration data
- Statistical thresholds are calculated using quantiles:
 - Weak threshold = 25th percentile
 - Strong threshold = 75th percentile

Purpose:

- Adapt system to individual player performance
- Improve classification accuracy

Pseudo Code: Threshold Calculation

LOAD calibration dataset

FOR each swing:
 compute power

weak_threshold = 25th percentile(power)
strong_threshold = 75th percentile(power)
SAVE thresholds

6.6 Feature Engineering

Feature engineering transforms raw data into meaningful parameters that improve model performance.

6.6.1 Primary Features

These are directly derived from sensor data:

- Speed → derived from angular velocity
- Impact → derived from acceleration magnitude
- Duration → time difference of swing

6.6.2 Derived Features

These features provide deeper insights:

- Power
 = Speed × Impact
- Efficiency
 = Impact / Duration
- Energy
 = Power × Duration

Pseudo Code: Feature Calculation

FOR each swing:
 power = speed * impact
 efficiency = impact / duration
 energy = power * duration

Code Concept Snippet

```
df["power"] = df["speed"] * df["impact"]  
df["efficiency"] = df["impact"] / (df["duration"] + 1e-6)  
df["energy"] = df["power"] * df["duration"]
```

6.7 Stroke Feature Engineering

Additional features are created specifically for stroke classification:

- Acceleration magnitude, Gyroscope magnitude, Peak acceleration, Peak angular velocity

These features help distinguish between different stroke types such as smash, drop, and clear.

6.8 Data Preprocessing

Before applying machine learning models, the dataset is cleaned and prepared.

Steps Involved:

- Removal of missing values, Selection of relevant features, Normalization (if required), Feature alignment for ML models

6.9 Importance of Feature Engineering

Feature engineering significantly improves system performance by:

- Enhancing classification accuracy, Reducing noise and irrelevant data, Enabling advanced analytics, Improving interpretability of results

CHAPTER 7

MACHINE LEARNING AND HYBRID MODEL DESIGN

7.1 Introduction

Machine learning plays a crucial role in transforming raw motion data into meaningful performance insights. In this project, machine learning models are used to classify badminton swings and identify stroke types based on extracted features.

In addition to machine learning, a rule-based approach is incorporated to enhance reliability. This combination forms a hybrid intelligence system, improving both accuracy and interpretability.

7.2 Overview of Machine Learning Approach

The system uses supervised learning techniques where models are trained on labeled datasets of badminton swings.

Key Tasks:

- Swing strength classification
- Stroke type classification

Input to Models:

- Engineered features such as speed, impact, power, and energy

Output:

- Swing category (WEAK, MEDIUM, STRONG)
- Stroke type (SMASH, DROP, CLEAR, DRIVE)

7.3 Swing Classification Model

This model classifies the intensity of each swing.

Classes:

- WEAK, MEDIUM, STRONG

Features Used:

- Speed, Impact, Power, Efficiency

Working:

The trained model analyzes input features and predicts the swing category based on learned patterns.

Pseudo Code: Swing Prediction

```
LOAD swing_model
```

```
FOR each swing:
```

```
    features = [speed, impact, power, efficiency]
```

```
    prediction = swing_model.predict(features)
```

7.4 Stroke Classification Model

This model identifies the type of badminton stroke.

Classes:

- SMASH
- DROP
- CLEAR
- DRIVE

Features Used:

- Acceleration magnitude
- Gyroscope magnitude

- Peak acceleration
- Energy

Working:

The model distinguishes stroke types based on motion patterns and feature variations.

Pseudo Code: Stroke Prediction

LOAD stroke_model

FOR each swing:

```
    features = [acc_mag, gyro_mag, peak_acc, energy]
    stroke = stroke_model.predict(features)
```

7.5 Model Training Process

The models are trained using labeled datasets collected during experimentation.

Steps Involved:

1. Data collection and labeling
2. Feature engineering
3. Splitting dataset into training and testing sets
4. Model training using appropriate algorithm
5. Performance evaluation

Model Storage:

- Trained models are saved using serialization (.pkl files)
- Loaded during prediction phase

7.6 Hybrid Model Design

A hybrid approach is implemented by combining machine learning predictions with rule-based classification.

Why Hybrid?

- Improves robustness
- Handles edge cases
- Provides interpretable logic

Rule-Based Classification

```
IF power < weak_threshold:  
    label = WEAK  
ELSE IF power < strong_threshold:  
    label = MEDIUM  
ELSE:  
    label = STRONG
```

Final Decision Logic

```
IF ML prediction is confident:  
    use ML result  
ELSE:  
    fallback to rule-based classification
```

7.7 Model Performance

The system achieves high classification accuracy based on experimental results.

Observed Performance:

- Accuracy $\approx$ 98% for swing classification
- Reliable stroke classification performance

Factors Contributing to Accuracy:

- Effective feature engineering
- Clean dataset
- Hybrid decision logic

7.8 Advantages of Machine Learning Approach

- Enables automatic pattern recognition
- Reduces dependency on manual thresholds
- Adapts to different player styles
- Supports scalable and extensible design

CHAPTER 8

SYSTEM IMPLEMENTATION AND ALGORITHM DESIGN

8.1 Introduction

This chapter describes the practical implementation of the Smart Badminton Performance Analytics System. It covers the integration of hardware,

software, machine learning models, and user interfaces. Additionally, the algorithms used for swing detection, feature extraction, and classification are explained.

8.2 Implementation Overview

The system is implemented in three main modules:

- Embedded Module (ESP32)
- Data Processing & Machine Learning Module (Python)
- User Interface Module (Web Dashboard & Streamlit)

Each module works independently and communicates through structured data.

8.3 Embedded System Implementation (ESP32)

The ESP32 firmware is responsible for real-time data acquisition, swing detection, and communication.

Key Tasks Implemented:

- Sensor initialization using I2C
- Continuous data reading from MPU6050
- Gravity filtering and signal processing
- Swing detection using threshold logic
- Data logging in CSV format
- Hosting a web server for real-time monitoring

Algorithm 1: Swing Detection Algorithm

The system detects a valid swing based on acceleration thresholds and time duration.

INPUT: Accelerometer and gyroscope data

WHILE system is running:

 read sensor data

 compute total acceleration

 IF totalAcceleration > START_THRESHOLD:

 start swing

 record start time

 WHILE swing is active:

 track peak speed and impact

 IF totalAcceleration < END_THRESHOLD AND duration >

 MIN_DURATION:

```
end swing
store swing data
```

8.4 Feature Extraction Implementation

After detecting a swing, key parameters are computed.

Algorithm 2: Feature Extraction

```
FOR each detected swing:
    speed = function(angular velocity)
    impact = function(acceleration)
    duration = end_time - start_time

    power = speed * impact
    efficiency = impact / duration
```

Code Concept

```
power = speed * impact
efficiency = impact / (duration + 1e-6)
```

8.5 Machine Learning Implementation

The system uses trained models stored as .pkl files.

Algorithm 3: Swing Classification

```
LOAD swing_model
FOR each swing:
    input_features = [speed, impact, power, efficiency]
    prediction = swing_model.predict(input_features)
```

Algorithm 4: Stroke Classification

```
LOAD stroke_model
FOR each swing:
    input_features = [acc_mag, gyro_mag, peak_acc, energy]
    stroke_type = stroke_model.predict(input_features)
```

8.6 Hybrid Classification Implementation

Algorithm 5: Rule-Based Classification

```
IF power < weak_threshold:
    label = WEAK
ELSE IF power < strong_threshold:
    label = MEDIUM
ELSE:
    label = STRONG
```

Final Decision Strategy

IF ML prediction is reliable:

 use ML result

ELSE:

 use rule-based result

CHAPTER 9

RESULTS AND PERFORMANCE ANALYSIS

9.1 Introduction

This chapter presents the experimental results obtained from the Smart Badminton Performance Analytics System. It evaluates system performance in terms of accuracy, real-time capability, and analytical insights generated from player data.

9.2 Experimental Setup

- Hardware: ESP32 + MPU6050 mounted on racket
- Modes Used: Calibration and Game mode
- Data Collected: Multiple swing sessions (event-based)
- Processing: Python pipeline with trained ML models
- Visualization: ESP32 Web Dashboard and Streamlit UI

9.3 Dataset Summary

- Total swings recorded: 102
- Features per swing: speed, impact, duration, power, efficiency
- Dataset type: event-based CSV (one row per swing)
- Calibration samples: 30 swings

Stroke Classification Performance

- Successfully classified strokes into:
 - SMASH, DROP, CLEAR, DRIVE

Observation:

- Distinct motion patterns enabled effective classification

- Feature engineering significantly improved accuracy

9.4 Real-Time System Performance

- Data acquisition delay: very low (~milliseconds)
- Dashboard refresh rate: ~200 ms
- Smooth real-time monitoring achieved

Observation:

- No significant lag during swing detection
- Stable performance during continuous sessions

9.5 Visualization Results

ESP32 Dashboard

- Displays real-time: Speed, Impact, Swing count

Streamlit Dashboard

- Displays: Performance metrics, Stroke distribution, AI suggestions, Graphical analysis

9.6 Comprehensive session report

SESSION OVERVIEW

Metric	Value
Total Swings	102
Weak Swings	19
Medium Swings	63
Strong Swings	20
Player Level	Needs Major Improvement
Player Type	Attacker

PERFORMANCE METRICS

Metric	Your Average	Pro Average	Gap
Speed	47.88	53.51	-5.63
Impact	34.72	47.35	-12.63
Power	1700.34	2532.58	-832.24
Efficiency	77.73	69.01	8.71

FATIGUE & CONSISTENCY

GAP ANALYSIS

Speed: Major deficit in swing acceleration

Impact: Weak shuttle contact force

Power: Very low explosive strength

Metric	Value
Fatigue Status	Fresh performance throughout session
Fatigue Score	-11.04%
Speed Drop	-9.65%
Impact Drop	-7.48%
Power Drop	-16.00%
Speed CV	0.156
Impact CV	0.169
Power CV	0.261
Consistency Score	80.47/100

Fig: 9.1 Comprehensive session report

9.10 System Strengths

- High classification accuracy
- Real-time processing capability
- Hybrid ML + rule-based reliability
- Advanced analytics and insights
- User-friendly visualization

CHAPTER 10

FUTURE SCOPE AND PRODUCTIZATION

10.1 Introduction

The current Smart Badminton Performance Analytics System has been developed as a functional prototype using ESP32 and MPU6050 for real-time motion tracking and machine learning-based performance analysis. While the system demonstrates strong accuracy and usability, it is primarily a lab-scale implementation.

10.2 Transition from Prototype to Product

The productization process focuses on reducing size and weight, improving power efficiency, and upgrading hardware components for wearable deployment inside a badminton racket.

10.3 Proposed Hardware Upgrades

Microcontroller Upgrade

- From ESP32 Dev Board, To: Ultra-low power compact boards such as:
 - Seeed Studio XIAO nRF52840, ESP32-C3

Sensor Upgrade

- From MPU6050, To: ICM42688 (high-performance IMU sensor) for Better accuracy, lower noise, improved stability

Power System Upgrade

- From standard battery setup
- To: 200mAh ultra-compact Li-Po battery

10.4 COMPARISON: PROTOTYPE vs FUTURE PRODUCT

Parameter	Current Prototype (ESP32 + MPU6050)	Future Product (XIAO nRF52840 / ESP32-C3 + ICM42688)
Microcontroller Weight	~25–30 g	~3–10 g
Sensor Weight	~4–5 g (MPU6050 module)	~2–3 g (ICM42688 module)
Battery Weight	~11–20 g	~4–6 g (200mAh Li-Po)
Total System Weight	~45–80 g	~10–20 g
Power Consumption	High (WiFi active mode)	Very Low (BLE / optimized sleep modes)
Communication	WiFi-based streaming	BLE (low power, mobile friendly)
Accuracy	Good (prototype level)	Higher (low-noise sensor + better IMU)

10.5 Key Improvements in Product Version

✓ Weight Reduction

- Nearly **80–90% reduction in total system weight**

✓ Power Efficiency

- BLE communication instead of continuous WiFi, Deep sleep modes for microcontroller, Ultra-low power IMU (ICM42688)

✓ Sensor Accuracy

- Reduced drift and noise, Better motion tracking resolution, Improved stroke classification reliability

10.6 Future Product Vision

The upgraded system can evolve into a:

- Smart badminton racket with embedded AI
- Wearable sports performance tracker
- Mobile-connected training ecosystem

PROJECT SUMMARY

The Smart Badminton Performance Analytics System is an intelligent solution designed to analyze player performance using embedded systems and machine learning techniques. The system integrates an ESP32 microcontroller with an MPU6050 inertial measurement unit (IMU) to capture real-time motion data such as acceleration and angular velocity during badminton swings. A threshold-based algorithm is used to detect valid swings, ensuring accurate event-based data collection.

Key parameters including speed, impact, and duration are extracted and further processed through feature engineering to derive advanced metrics such as power, efficiency, and energy. These features are used as inputs for machine learning models that classify swings into WEAK, MEDIUM, and STRONG categories, and identify stroke types such as SMASH, DROP, CLEAR, and DRIVE. A hybrid approach combining machine learning with rule-based classification improves overall system reliability and interpretability.

The processed data is analyzed to generate meaningful insights including player performance level, consistency score, fatigue detection, and comparison with professional benchmarks. The system provides real-time feedback through a web-based dashboard and a Streamlit interface, enabling effective visualization of performance metrics.

The project demonstrates high accuracy, efficient real-time processing, and a scalable architecture. Furthermore, it outlines future enhancements for product development, including the use of low-power microcontrollers, advanced sensors, and compact battery systems to enable a lightweight, wearable smart sports device for practical deployment.

